BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT30** **Date: 24-01-2026**

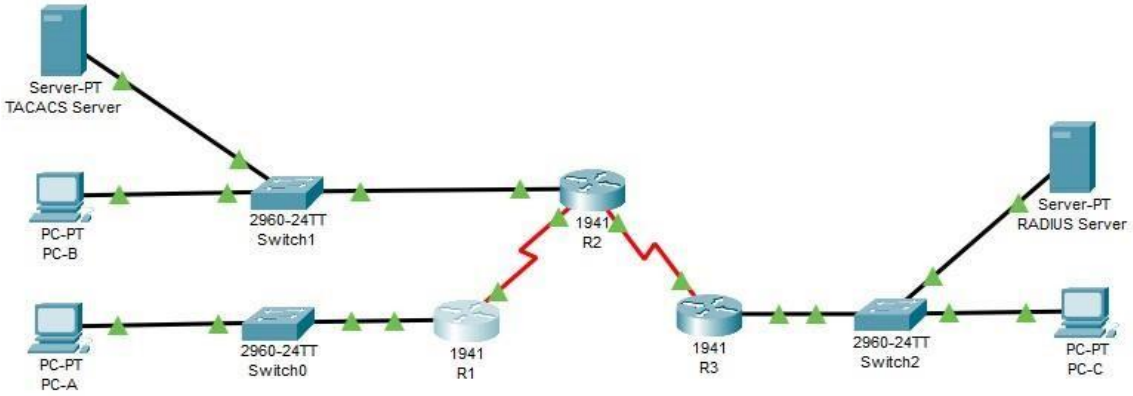
Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 4A

Q) Configure AAA Authentication: Configure a local user account on the Router and configure authentication on the console and vty lines using local AAA.

Steps:

1. Open cisco packet tracer → open new page.
2. Drag and drop 3 PC, 3 switch(2960), 2 server, 3 router(1941) → change devices names and all routers change hostname as will → then click on router → physical → zoom in → turn off switch then drag and drop HWIC-2T to both box → close.
3. Wire connections: Switch-A-R1(g0/1), Switch-B-R2(g0/0), Switch-C-R3(g0/1) (copper straighttthrought wire for connection). R1-R2(s0/0/0) R2-R3(0/0/1)(serial DCE wire for connection). Rest of all same use copper straight-throught wire for connection.



4. Assign the ip address to all:

IP Config	IPv4 Address	Subnet Mask	Default Gateway
R1	192.168.1.1	255.255.255.0	-
R2	192.168.2.1	255.255.255.0	-
R3	192.168.3.1	255.255.255.0	-
R1 S(0/0/0)	10.1.1.1	255.255.255.252	-
R2 S(0/0/0)	10.1.1.2	255.255.255.252	-
R2 S(0/0/1)	10.2.2.1	255.255.255.252	-
R3 S(0/0/1)	10.2.2.2	255.255.255.252	-
TACACS Server	192.168.2.2	255.255.255.0	192.168.2.1
RADIUS Server	192.168.3.2	255.255.255.0	192.168.3.1
PC-A	192.168.1.3	255.255.255.0	-
PC-B	192.168.2.3	255.255.255.0	-
PC-C	192.168.3.3	255.255.255.0	-

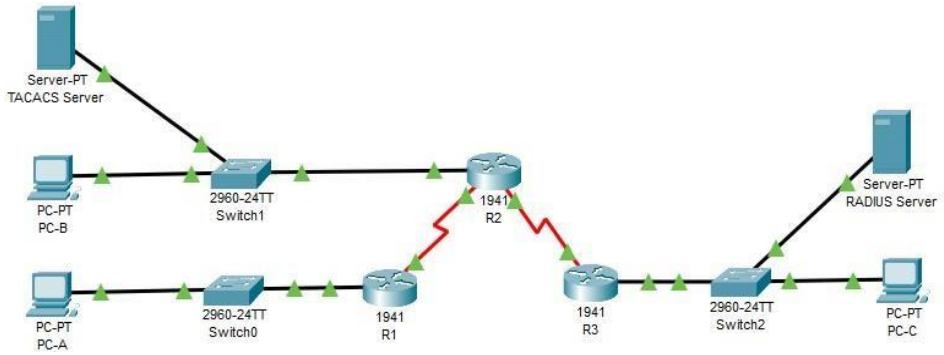
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT30** **Date: 24-01-2026**

Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 4A



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC-A	R1	ICMP		0.000	N	0	(ec
	Successful	PC-B	R2	ICMP		0.000	N	1	(ec
	Successful	TACA...	R2	ICMP		0.000	N	2	(ec

5. Click on R1→CLI.
   - username Admin1 secret adminpa55
   - aaa new-model
   - aaa authentication login default local
   - line console 0
   - login authentication default
   - end
   - exit

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#username Admin1 secret adminpa55
R1(config)#aaa new-model
R1(config)#aaa authentication login default local
R1(config)#line console 0
R1(config-line)#login authentication default
R1(config-line)#end
R1#exit
```

   - Enter username:Admin1 password: adminpa55

```
User Access Verification

Username: Admin1
Password:
R1>
```

   - ip domain-name ccnasecurity.com
   - crypto key generate rsa
   - 1024

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT30** **Date: 24-01-2026**

Course Name: Cybersphere and Security(CSS) **Page no:**

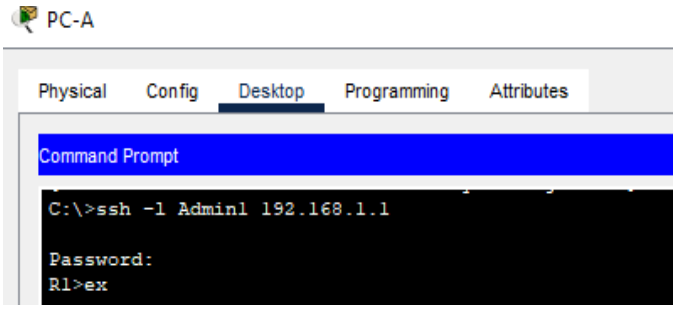
Practical Number:- 4A

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip domain-name ccnasecurity.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#
```

- aaa authentication login SSH-LOGIN local
- line vty 0 4
- login authentication SSH-LOGIN
- transport input ssh
- end

```
R1(config)#aaa authentication login SSH-LOGIN local
*Mar 1 1:15:13.146: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

6. Click on PC-A à desktop à command prompt.
   - Shh –l Admin1 192.168.1.1



Teacher's Signature